

January 2026 CSE 314: Online 1 (A1)

Time: 25 minutes

During the battle for Westeros, raven messages from different regions have become mixed throughout the archive. Each message may contain the following keywords:

```
winter
dragon
throne
```

Command

```
./online-2205XXX.sh <input_dir> <output_dir>
```

Requirements

1. Create `output_dir` if it does not already exist. Then create the following three subdirectories inside it:

```
output_dir/
|-- Stark/
|-- Targaryen/
|-- Royal/
```

2. Search through `input_dir` and all of its subdirectories.
3. For every file, calculate the following values using case-insensitive keyword matching:
 - `winter_count`: number of lines containing `winter`
 - `dragon_count`: number of lines containing `dragon`
 - `throne_count`: number of lines containing `throne`
 - `total_score`: sum of the three counts
4. Append `total_score` to the end of the file's complete original filename, including after its extension. For example, if `warning.msg` has a total score of 4, its new name will be:

```
warning.msg_4
```

5. Move every file according to the keyword with the highest count:
 - Move it to `output_dir/Stark/` if `winter_count` is the highest.
 - Move it to `output_dir/Targaryen/` if `dragon_count` is the highest.
 - Move it to `output_dir/Royal/` if `throne_count` is the highest.

6. After all files have been moved to their appropriate directories, add a numerical prefix to each file according to its position in the file-size order. Numbering must begin independently from 0 inside each category directory. For example:

```
0_warning.msg_4
1_prophecy.txt_6
2_battle.log_9
```

Assumptions

You may assume that:

- All input files are regular readable files.
- Every file has exactly one keyword whose count is strictly greater than the other two keyword counts. Therefore, classification ties will not occur.
- No line in any input file contains a keyword more than once.
- No two files moved in the same category directory have the same file size.
- Every input file has a unique filename, even when the files are in different subdirectories.
- `input_dir` and `output_dir` refer to different directory trees.

Sample Input

```
input_dir/
|-- north/
|   '-- warning.msg
|-- east/
|   |-- prophecy.log
|   '-- dragon_report.data
'-- capital/
    '-- royal_order.txt
```

Sample Output

Assuming `prophecy.log_4` is smaller than `dragon_report.data_3`, the output is:

```
output_dir/
|-- Stark/
|   '-- 0_warning.msg_3
|-- Targaryen/
|   |-- 0_prophecy.log_4
|   '-- 1_dragon_report.data_3
'-- Royal/
    '-- 0_royal_order.txt_4
```